AOA Important Repeated 2-Mark Questions and Answers (1-50)

1. Define Dynamic Programming.

Dynamic Programming is a method of solving problems by dividing them into overlapping subproblems and storing their solutions to avoid repeated computation.

2. Define Greedy Method.

Greedy Method is an algorithmic technique that builds a solution step-by-step by selecting the locally optimal choice at each stage.

3. Differentiate between Floyd's and Warshall's Algorithm.

Floyd's Algorithm finds shortest paths, whereas Warshall's Algorithm finds transitive closure.

4. Differentiate between Greedy and Dynamic Programming.

Greedy takes local optimal choices while Dynamic Programming solves and stores all subproblems.

5. Define Binary (0/1) Knapsack.

In 0/1 Knapsack problem, each item is either completely selected or completely rejected to maximize profit.

6. Differentiate between Prim's and Kruskal's Algorithm.

Prim's is vertex based whereas Kruskal's is edge based.

7. Name the single source shortest path algorithm.

Dijkstra's Algorithm.

8. Write the formula to calculate maximum profit using 0/1 Knapsack.

P[i,w] = max(P[i–1,w], pi + P[i–1,w–wi])

9. Name the method used for data compression.

Huffman Coding.

10. Define Backtracking.

Backtracking is a technique that builds solutions step-by-step and abandons non-promising solutions.

11. Define Branch and Bound.

Branch and Bound systematically explores solution spaces and eliminates non-promising branches using bounds.

12. Differentiate between Backtracking and Branch & Bound.

Backtracking is used for feasibility problems while Branch & Bound is used for optimization problems.

13. Write the formula for upper bound in 0/1 Knapsack.

UB = current profit + remaining capacity × (profit/weight)

14. Define P Problems.

P problems can be solved in polynomial time using deterministic algorithms.

15. Define NP Problems.

NP problems are problems whose solutions can be verified in polynomial time.

16. Define NP-Complete Problems.

NP-Complete problems are the hardest problems in NP.

17. Define NP-Hard Problems.

NP-Hard problems are at least as hard as NP-Complete problems.

18. Define Horspool Algorithm.

Horspool Algorithm is a string matching algorithm using shift tables.

19. Define Shift Table.

A shift table stores shift values used during string matching.

20. Name the method used for data compression.

Huffman Coding.

21. Differentiate between NP-Complete and NP-Hard Problems.

NP-Complete belongs to NP while NP-Hard may or may not belong to NP.

22. Define Space-Time Tradeoff.

Space-Time Tradeoff uses extra memory to reduce execution time or vice versa.

23. What is a Hash Table?

A Hash Table stores data using hash functions.

24. Define Collision in Hashing.

Collision occurs when multiple keys map to the same location.

25. Define Horspool String Matching Algorithm.

It compares pattern characters from right to left using shift tables.

26. Define Boyer–Moore Algorithm.

It uses bad character and good suffix heuristics for string matching.

27. List the four cases in Horspool Algorithm.

Character absent, present once, present multiple times, last character mismatch.

28. List the types in Boyer–Moore Algorithm.

Bad Character Heuristic and Good Suffix Heuristic.

29. Define Shift Table.

A shift table stores the number of shifts after mismatch.

30. Define Lower Bound.

Lower Bound is the minimum possible time required by any algorithm.

31. Define Computational Complexity.

It studies time and memory required by algorithms.

32. What is an Optimal Solution?

An optimal solution is the best feasible solution.

33. Define Objective Function.

Objective function measures the quality of a solution.

34. Define Feasibility Function in Greedy Method.

It checks whether a selected candidate satisfies constraints.

35. Define Safe Edge in MST.

A safe edge can be added to MST without forming a cycle.

36. Define Relaxation in Dijkstra's Algorithm.

Relaxation updates shortest distance when a shorter path is found.

37. Define Min-Heap.

A Min-Heap is a complete binary tree where parent value is smaller than child values.

38. What is the Time Complexity of Floyd's Algorithm?

O(n^3)

39. What is the Time Complexity of Warshall's Algorithm?

O(n^3)

40. Define Prefix-Free Code.

No codeword is the prefix of another codeword.

41. Define Huffman Coding.

Huffman Coding is a greedy method for lossless data compression.

42. What is Transitive Closure?

It shows whether a path exists between every pair of vertices.

43. Define State Space Tree.

It represents all possible states or solutions.

44. Define Hamiltonian Circuit.

A cycle visiting every vertex exactly once and returning to start.

45. Define Travelling Salesman Problem.

Find the minimum cost tour visiting all cities exactly once.

46. Define Sum of Subsets Problem.

Find subsets whose sum equals a target value.

47. Define Branching in Branch and Bound.

Branching divides a problem into smaller subproblems.

48. Define Bounding Function.

It calculates bounds to eliminate non-promising nodes.

49. What is an Adjacency Matrix?

A matrix representation of a graph.

50. Define Decision Tree.

A tree structure representing comparisons or decisions.